Higher Diploma in Science in Computer Science

Overview of Structure & Purpose of the Online Materials

Tutors

Slack

Moodle

Youtube

4 Key Web
Services

Tutors } Github ID

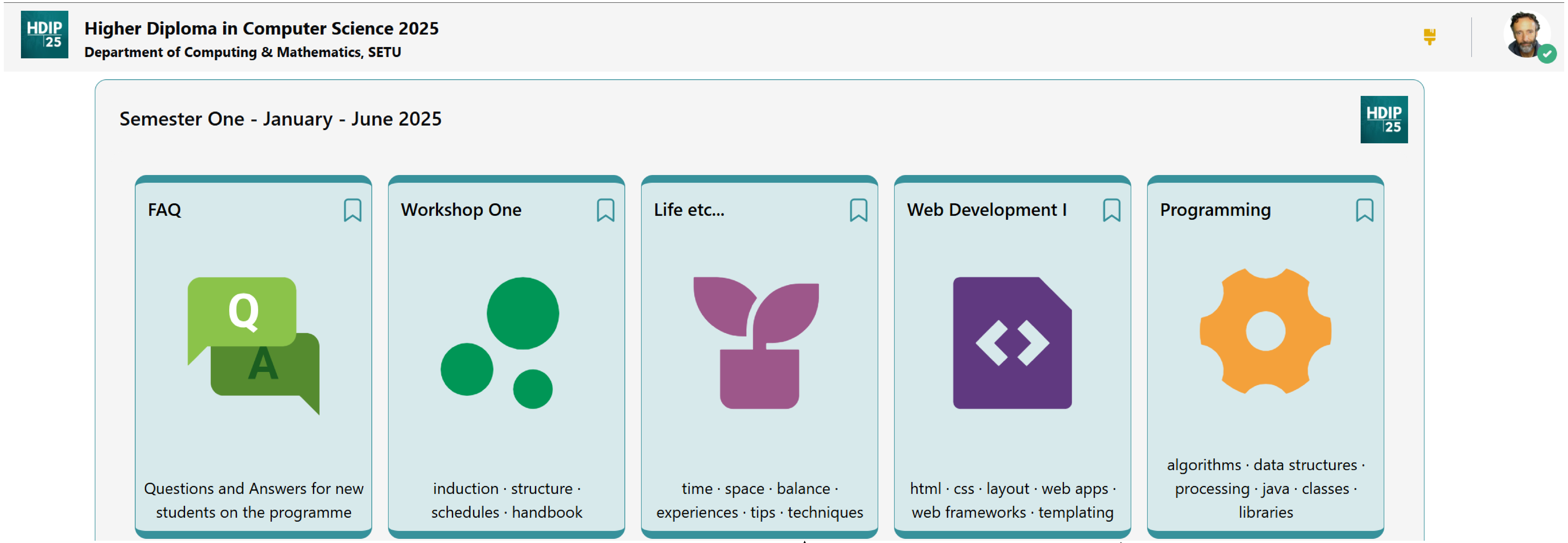
Slack } personal email

Moodle } wit email

Youtube } open

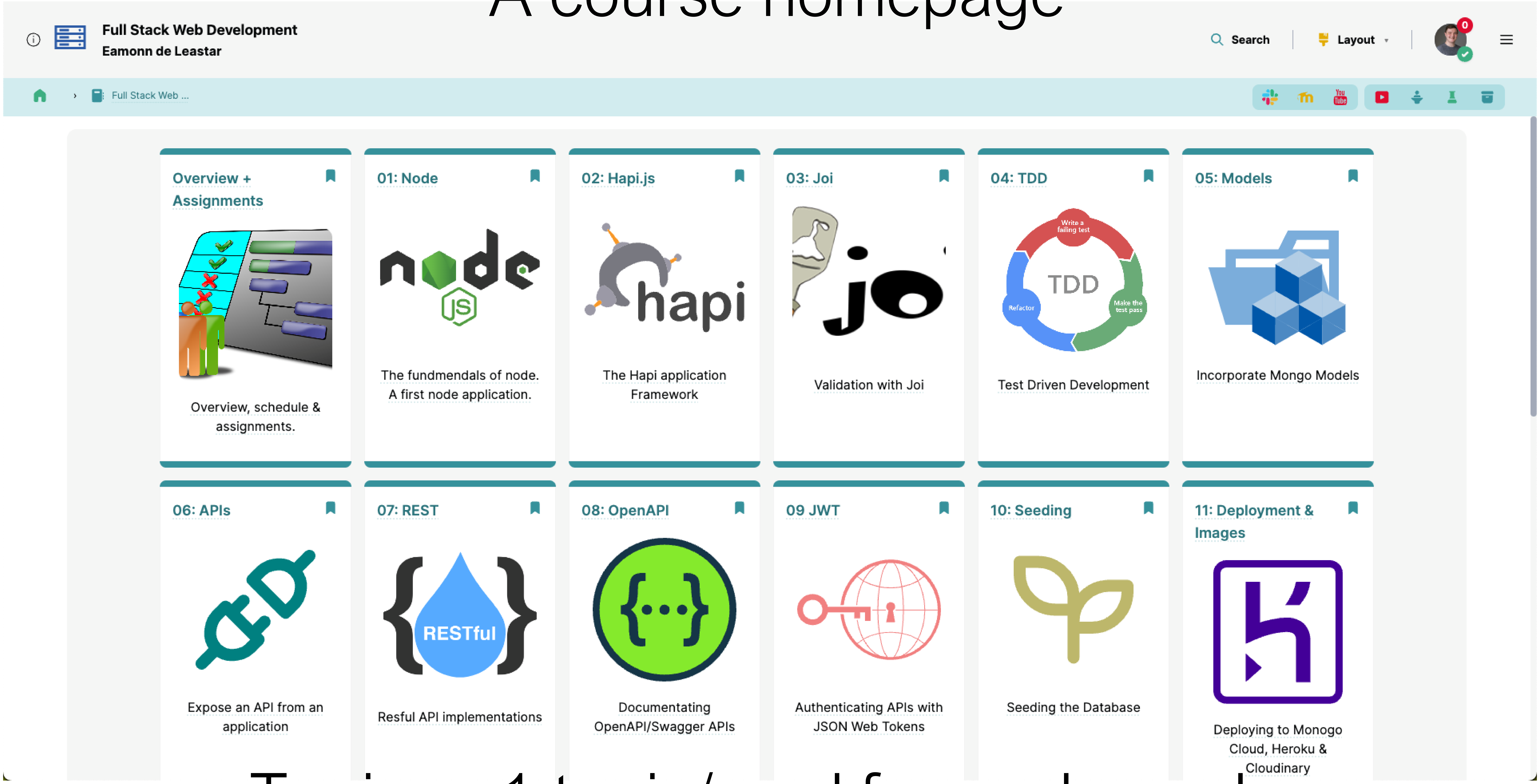
Programme homepage

https://tutors.dev/course/setu-hdip-comp-sci-2025



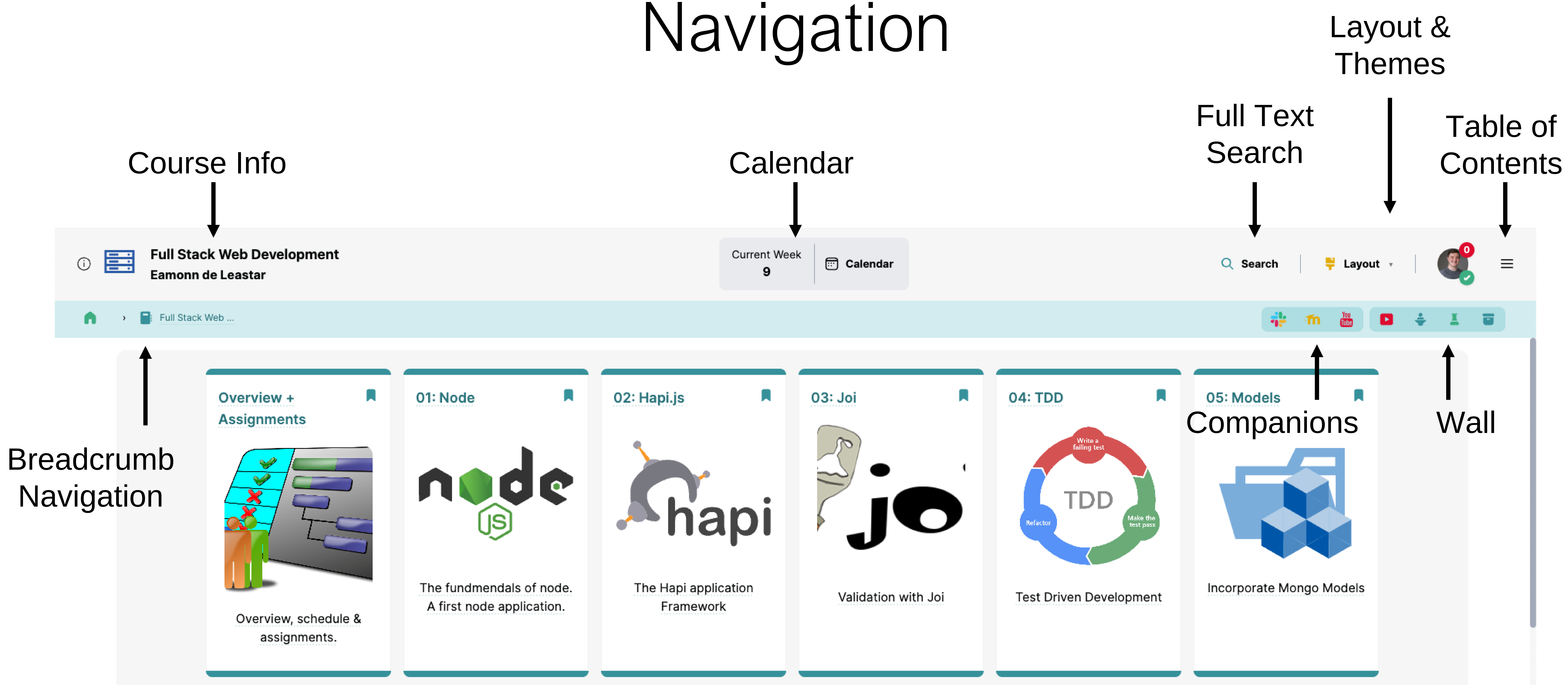
Links to individual module Home pages

A course homepage



Topics - 1 topic/card for each week

Navigation



Layout &
Themes
Full Text
Search
Table of
Contents
Course Info
Calendar
Full Stack Web Development
Eamonn de Leastar
Current Week
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Calendar
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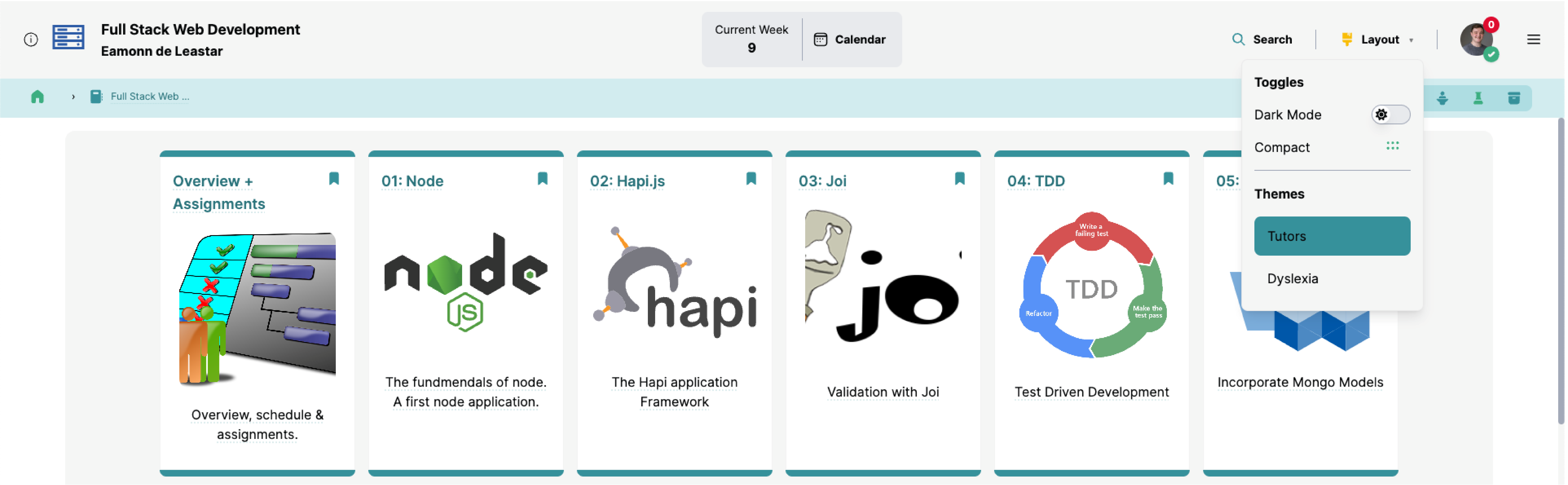
Full Stack Web ...
Overview +
Assignments
Overview, schedule &
assignments.
01: Node
The fundmendals of node.
A first node application.
02: Hapi.js
The Hapi application
Framework
03: Joi
Validation with Joi
04: TDD
Test Driven Development
05: Models
Incorporate Mongo Models
Companions
Wall
Breadcrumb
Navigation

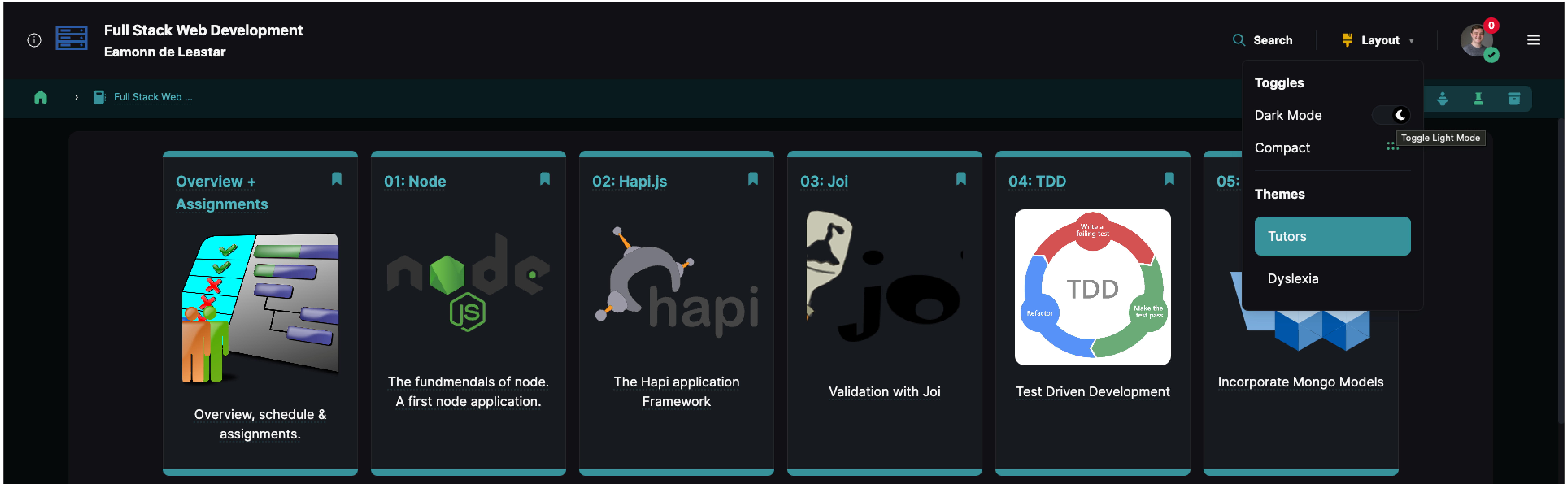
Navigation

Layout &
Themes



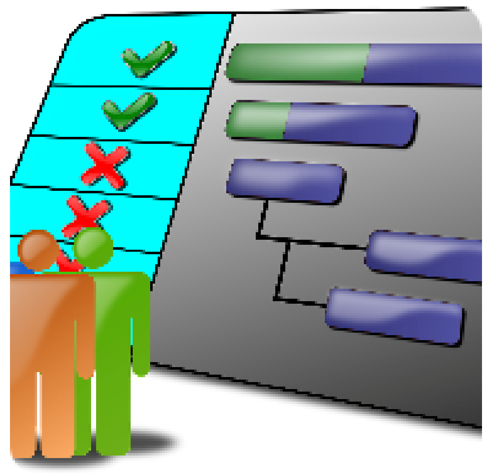
Theme Support

Layout &
Themes



Topics - 1 topic/card for each week

Overview + Assignments



Overview, schedule & assignments.

01: Node

The fundmendals of node. A first node application.

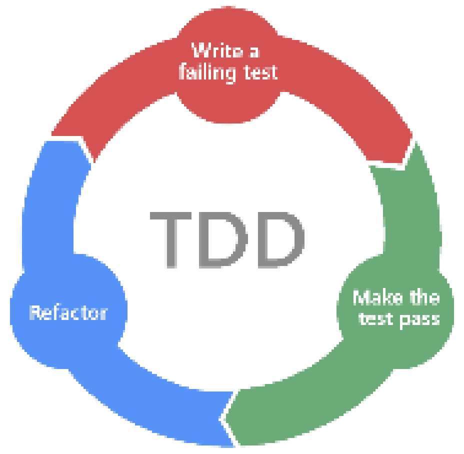
02: Hapi.js

The Hapi application Framework

03: Joi

Validation with Joi

04: TDD



Test Driven Development

05: Models

Incorporate Mongo Models

06: APIs

Expose an API from an application

07: REST

Resful API implementations

08: OpenAPI

Documentating OpenAPI/Swagger APIs

09 JWT

Authenticating APIs with JSON Web Tokens

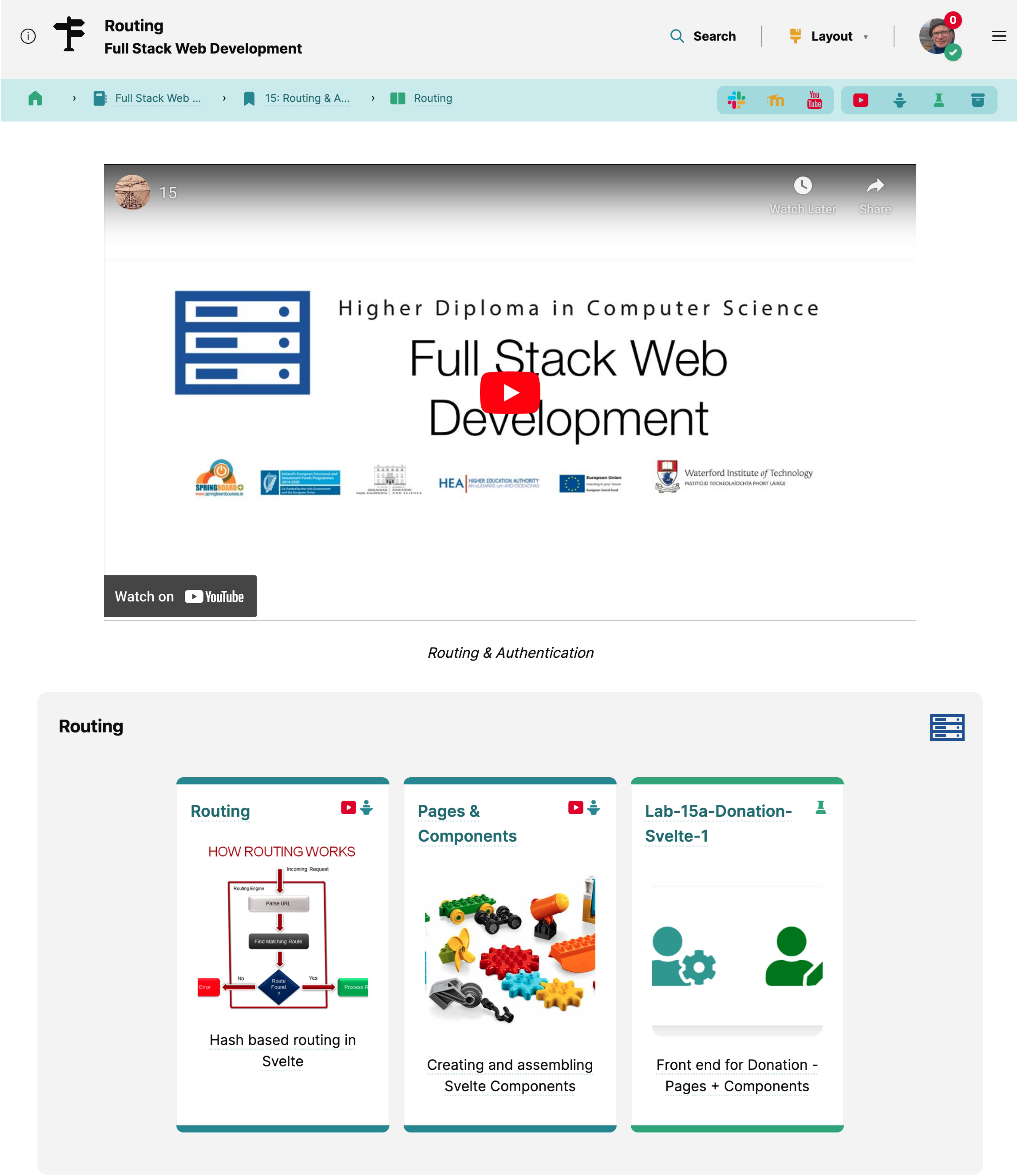
10: Seeding

Seeding the Database

11: Deployment & Images

Deploying to Monogo Cloud, Heroku & Cloudinary

Single Topic

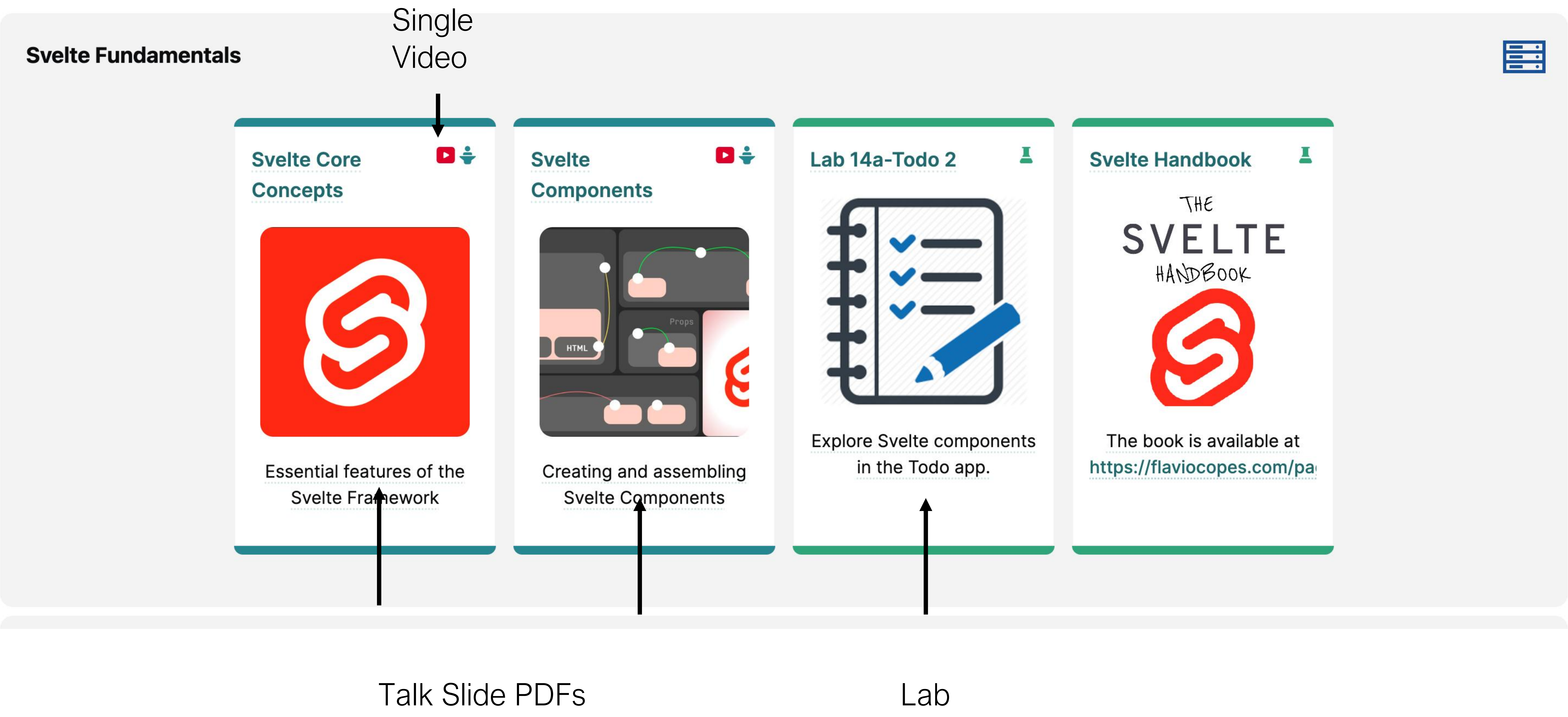


Topic Video

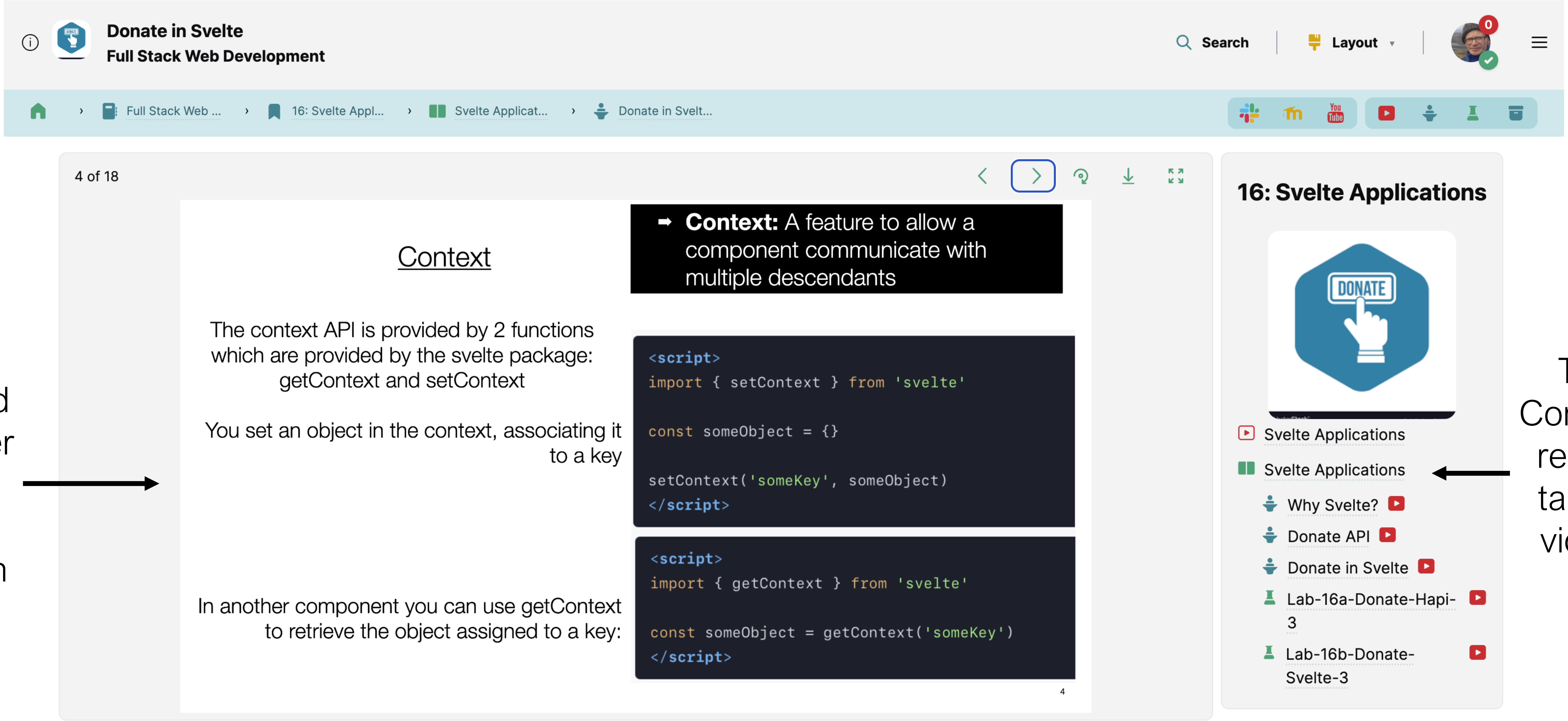
Pre-recorded or live stream

Sub Topic 1

Sub Topic


Single Video
Svelte Fundamentals
Svelte Core Concepts
Essential features of the Svelte Framework
Svelte Components
Props
HTML
Creating and assembling Svelte Components
Lab 14a-Todo 2
Explore Svelte components in the Todo app.
Svelte Handbook
THE SVELTE HANDBOOK
The book is available at https://flaviocopes.com/pa
Talk Slide PDFs
Lab

Embedded PDF reader custom with Navigation



Talk Context - related talks & videos

Talk

Labs

Lab Steps

Markdown formatted text

Code block syntax highlighting

- Objectives
- Create
- **Router**
- Components
- Pages
- Donate & Report
- Login & Signup
- Solutions

Router

Like web apps, front end applications also have the concept of a router. Svelte itself does not provide one, but there are several popular options:

- https://www.routify.dev/
- https://github.com/AlexxNB/tinro
- https://github.com/ItalyPaleAle/svelte-spa-router

We will use the last one - which is documented in detail in this svelte text book:

- Svelte 3 Up & Running

Start by installing it:

```
npm install svelte-spa-router
```

To try it out, create three svelte file in a new **pages** folder in the project:

pages/Main.svelte

```
<div class="notification is-link">
  Hello from Main!
</div>
```

pages/Login.svelte

```
<div class="notification is-info">
  Hello from Login!
</div>
```

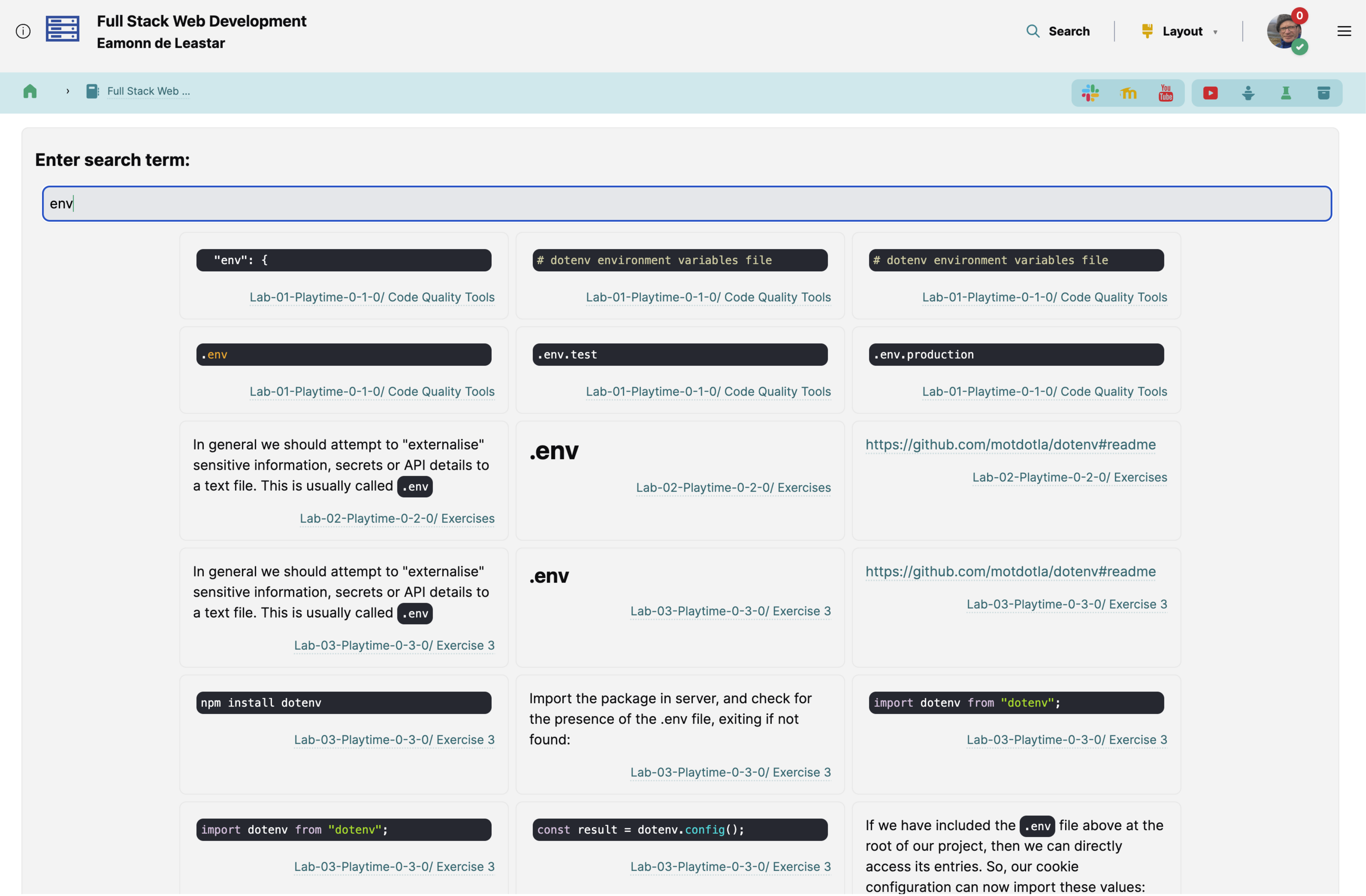
pages/Signup.svelte

```
<div class="notification is-warning">
  Hello from signup!
</div>
```

This is a revised App.svelte, to include the routing table:

App.svelte

Search


Full Stack Web Development
Eamonn de Leastar
Search
Layout
0
Full Stack Web ...
Enter search term:
env
"env": {
Lab-01-Playtime-0-1-0/ Code Quality Tools
dotenv environment variables file
Lab-01-Playtime-0-1-0/ Code Quality Tools
dotenv environment variables file
Lab-01-Playtime-0-1-0/ Code Quality Tools
.env
Lab-01-Playtime-0-1-0/ Code Quality Tools
.env.test
Lab-01-Playtime-0-1-0/ Code Quality Tools
.env.production
Lab-01-Playtime-0-1-0/ Code Quality Tools
In general we should attempt to "externalise" sensitive information, secrets or API details to a text file. This is usually called .env
Lab-02-Playtime-0-2-0/ Exercises
.env
Lab-02-Playtime-0-2-0/ Exercises
https://github.com/motdotla/dotenv#readme
Lab-02-Playtime-0-2-0/ Exercises
In general we should attempt to "externalise" sensitive information, secrets or API details to a text file. This is usually called .env
Lab-03-Playtime-0-3-0/ Exercise 3
.env
Lab-03-Playtime-0-3-0/ Exercise 3
https://github.com/motdotla/dotenv#readme
Lab-03-Playtime-0-3-0/ Exercise 3
npm install dotenv
Lab-03-Playtime-0-3-0/ Exercise 3
Import the package in server, and check for the presence of the .env file, exiting if not found:
Lab-03-Playtime-0-3-0/ Exercise 3
import dotenv from "dotenv";
Lab-03-Playtime-0-3-0/ Exercise 3
import dotenv from "dotenv";
Lab-03-Playtime-0-3-0/ Exercise 3
const result = dotenv.config();
Lab-03-Playtime-0-3-0/ Exercise 3
If we have included the .env file above at the root of our project, then we can directly access its entries. So, our cookie configuration can now import these values:

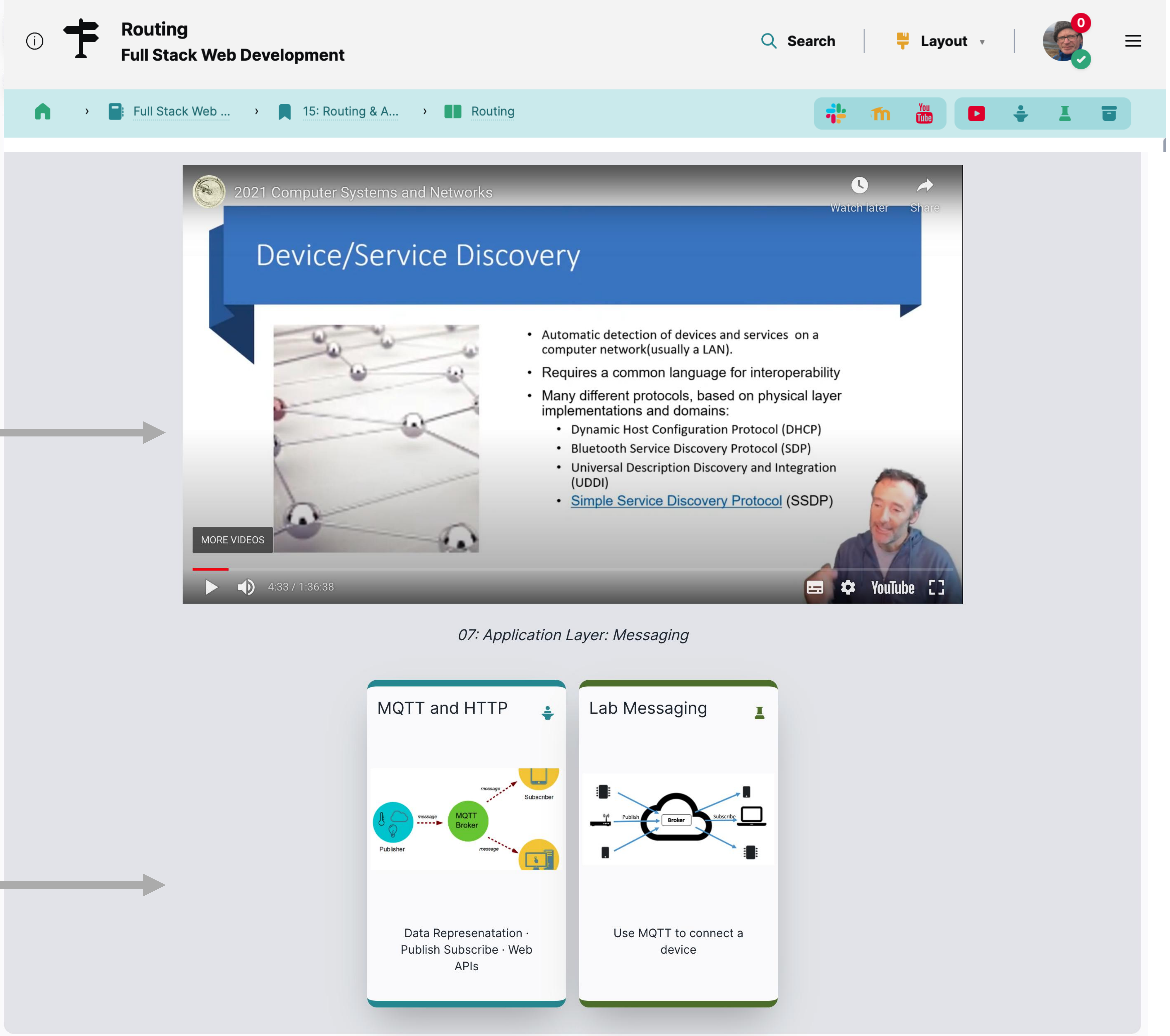
Youtube Live

Live Webinars Delivery

Live Stream

Live stream of webinar for given topic

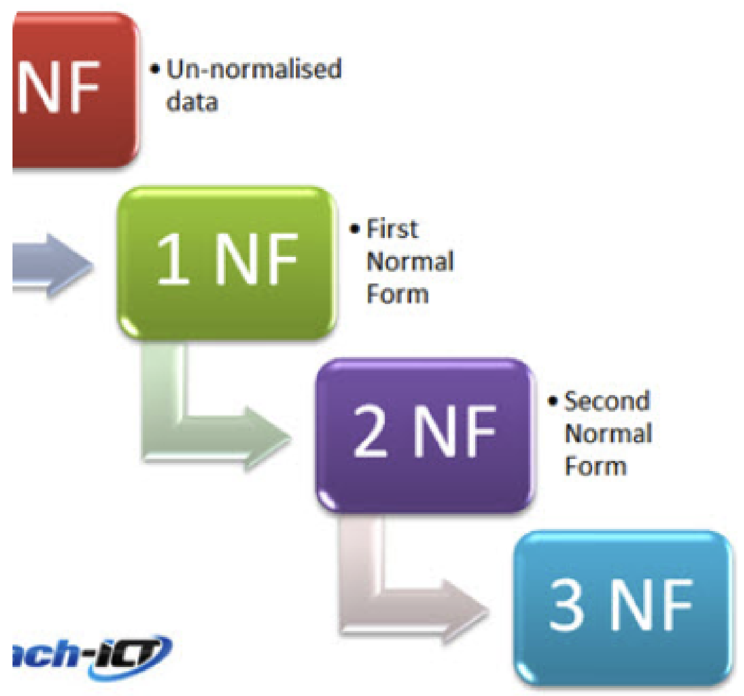
Slides & Labs for the topic



Pre-recorded Videos for selected slides & labs

Stages of Normalisation

The Process of Normalisation



In this lecture you will look at the Process of Normalisation.

First Normal Form

In this lecture you will look at First Normal Form.

Second Normal Form

In this lecture you will look at Second Normal Form.

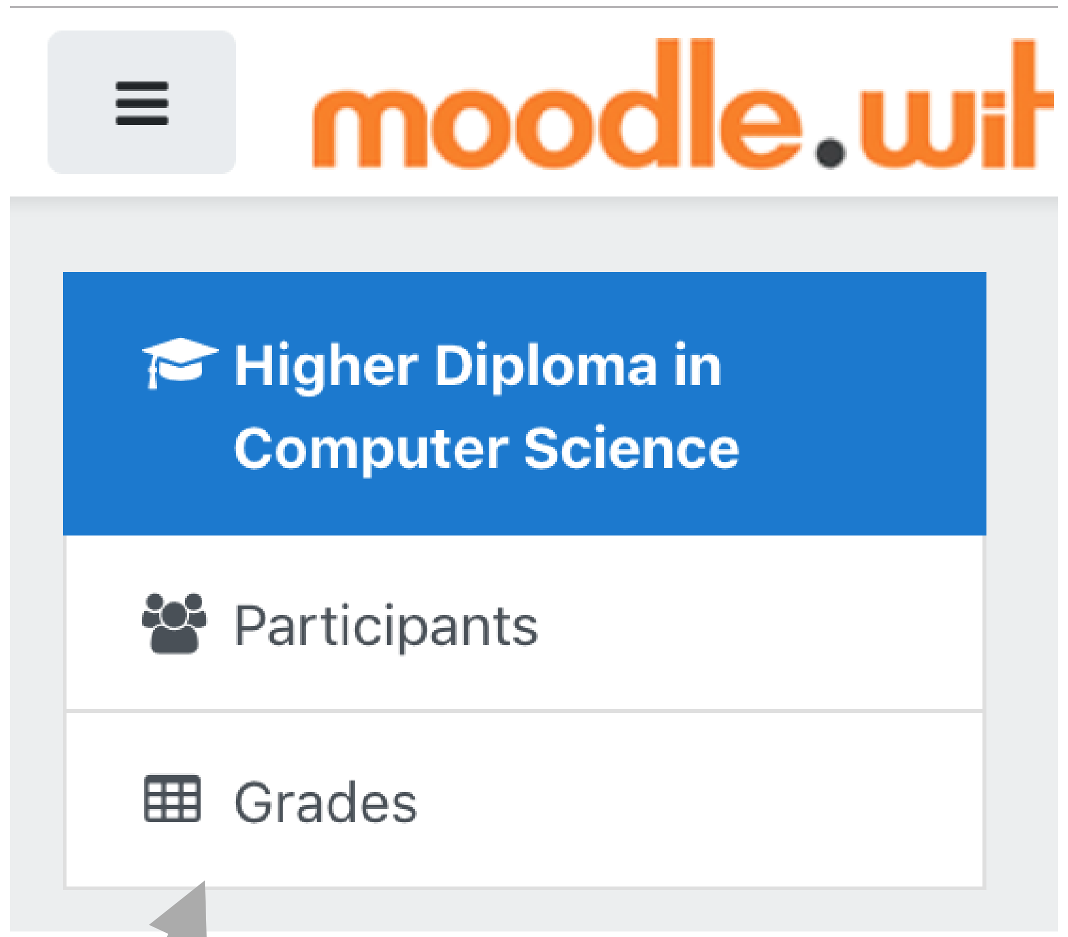
Third Normal Form

In this lecture you will look at Third Normal Form.

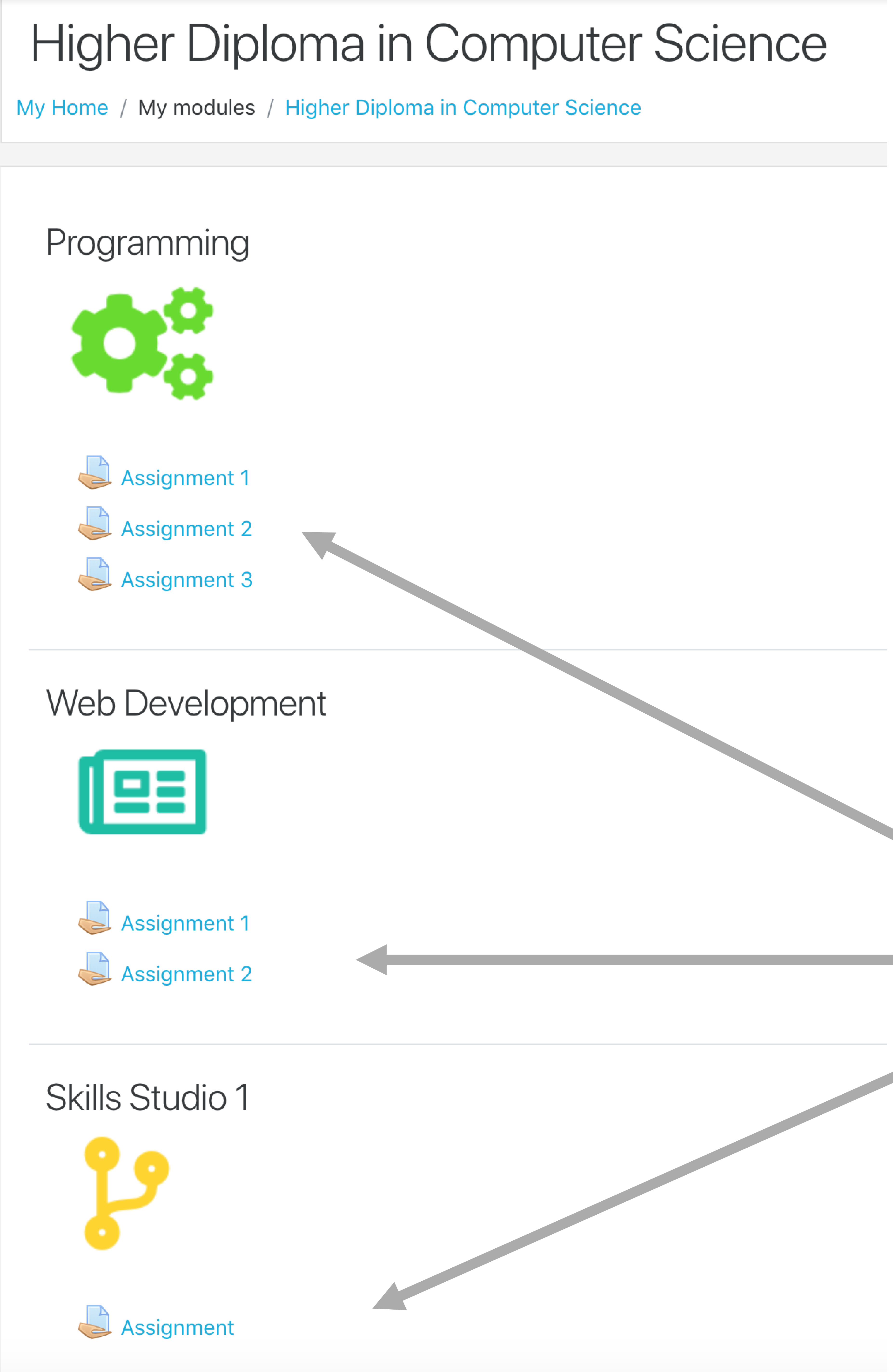
Moodle

WIT E-Learning Platform
Used for Assignment Submission + feedback + grades

Occasionally used for additional document distribution



Grades



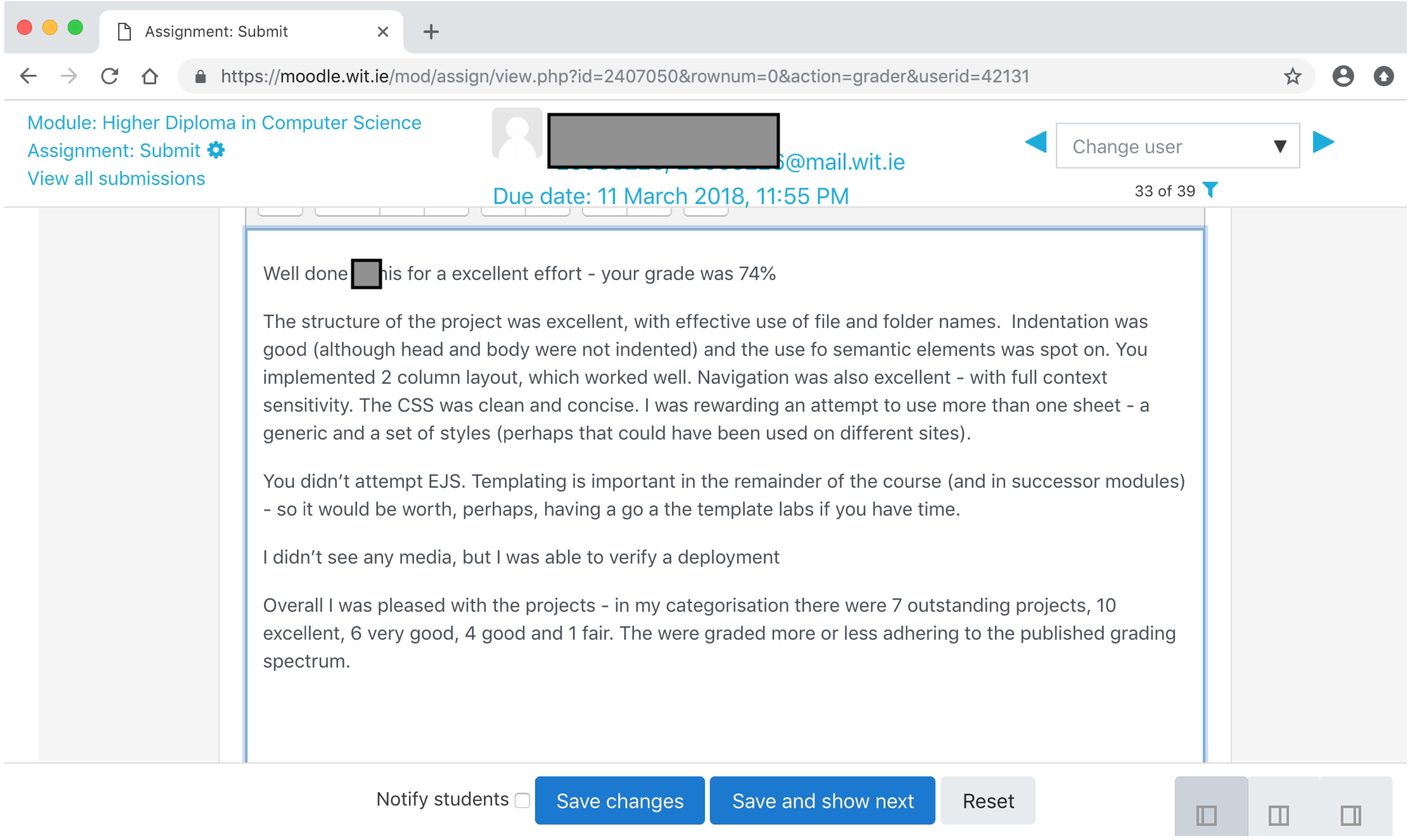
Assignment Submissions

Assessment feedback

Web Development

Assignment 1

Assignment 2

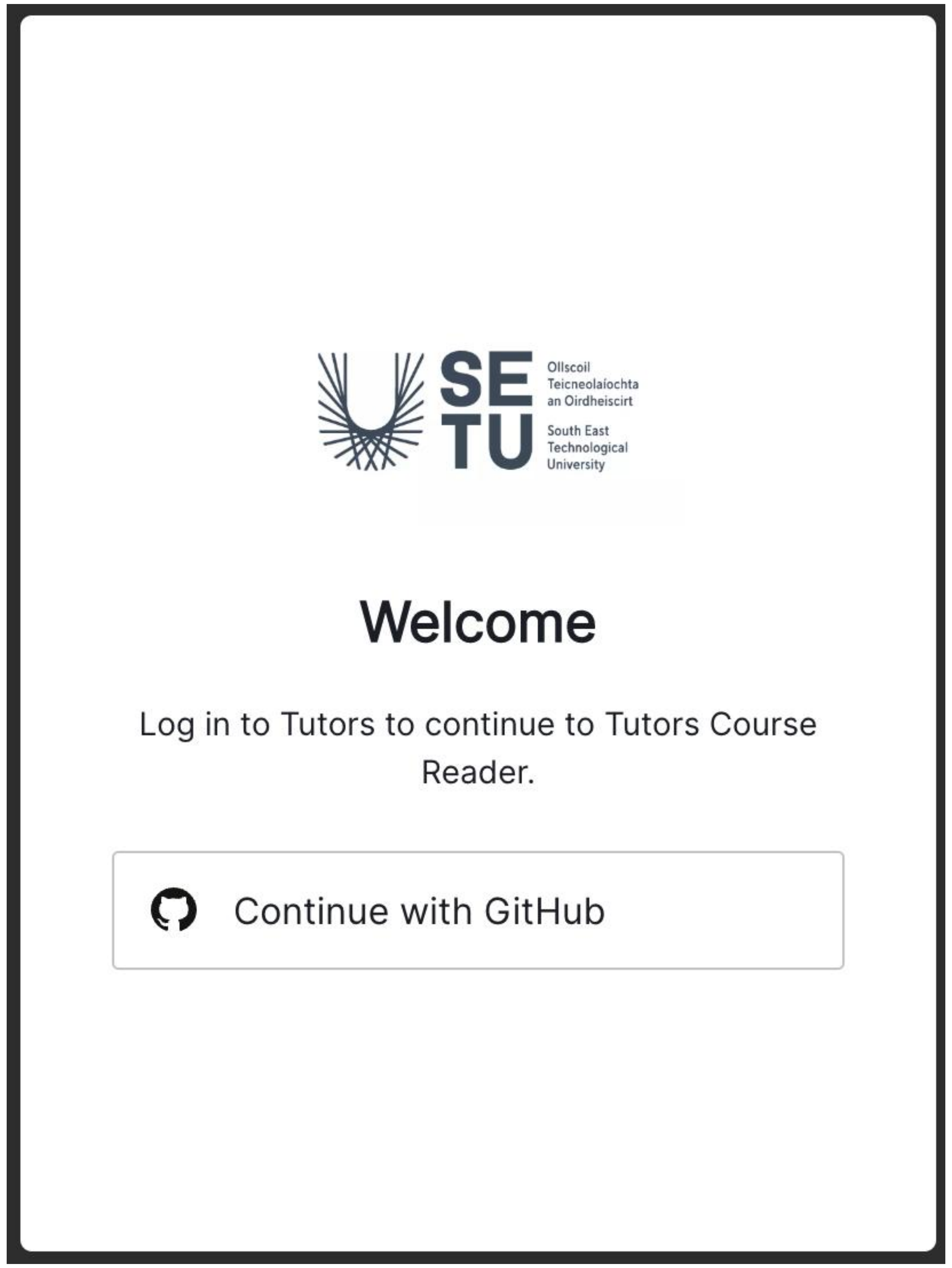


Tutors

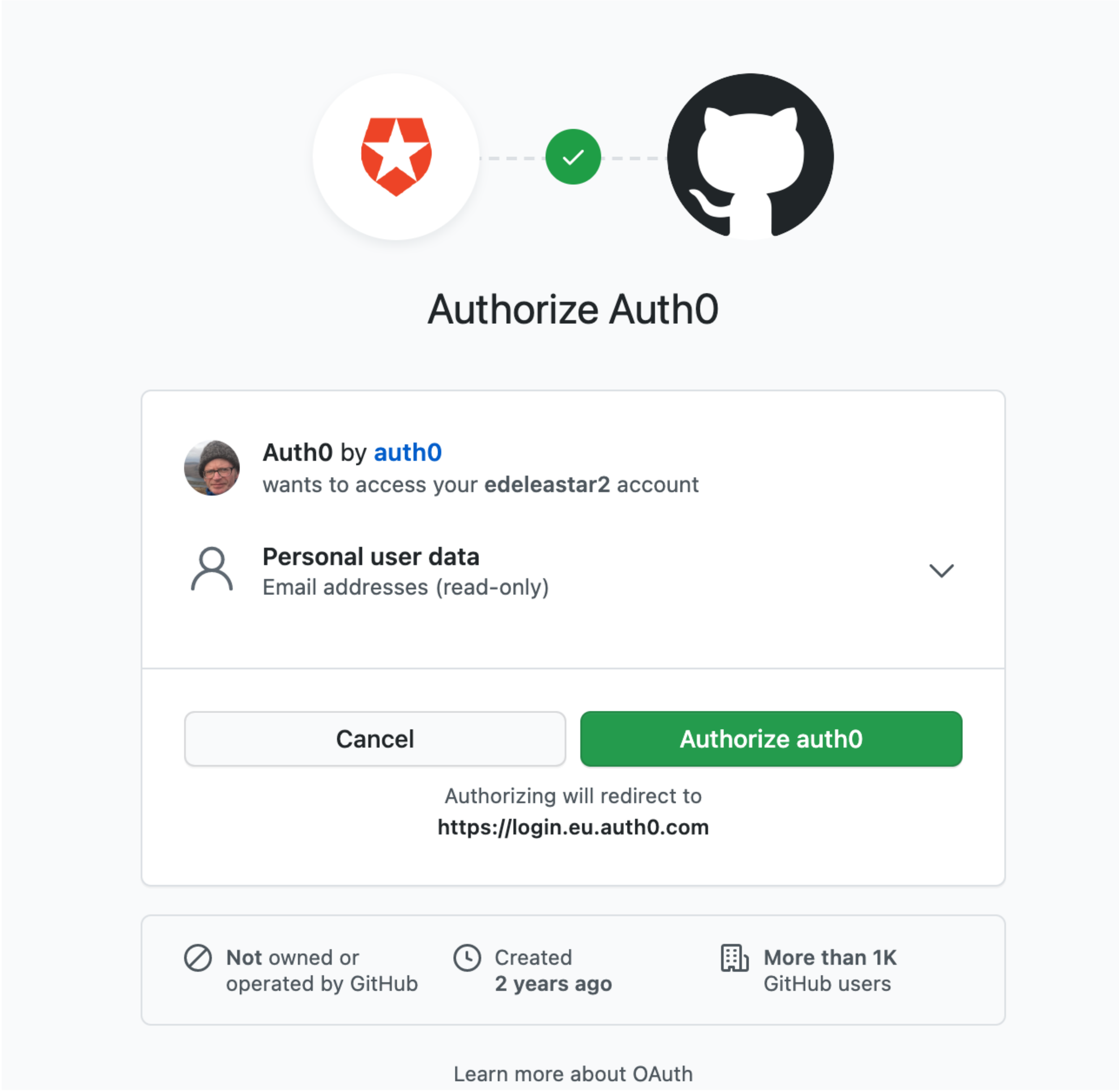
TutorsTime

TutorsLive

When accessing module, you will be asked to sign in with Github



… first time you will have to authorise …



TutorsTime

You have been authenticated via your GitHub credentials and are about to enter a Tutors course. Tutors will record how much time each view is active and send this to the TutorsTime data store. You can view this information via the TutorsTime feature on the course Navigator.

This data is available to you + the instructor of your course, but not to other students. No other data is gathered, nor is this data transmitted anywhere other than the TutorsTime data store.

TutorsLive also uses this data to display of panel of students currently online. You can opt out of this feature from the TutorsLive page.

To learn more about these features please consult:

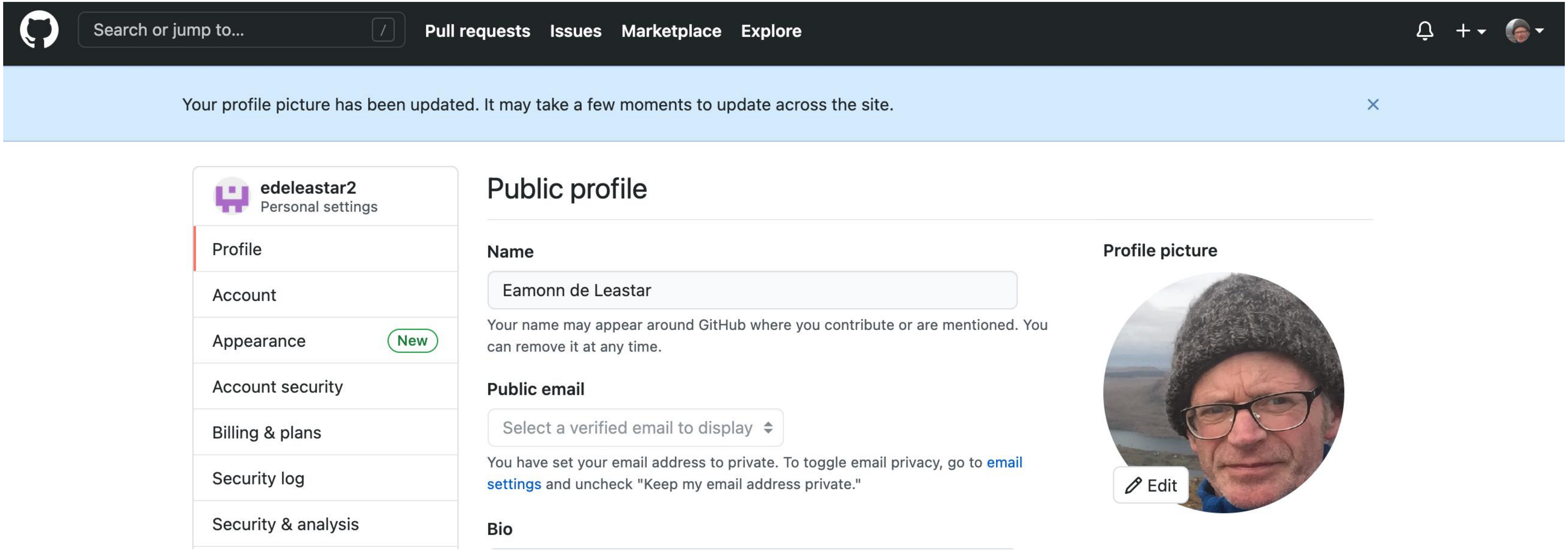
- Tutors Time FAQ
- Tutors Live FAQ

Tutors is an open source application - the data collection component is here.

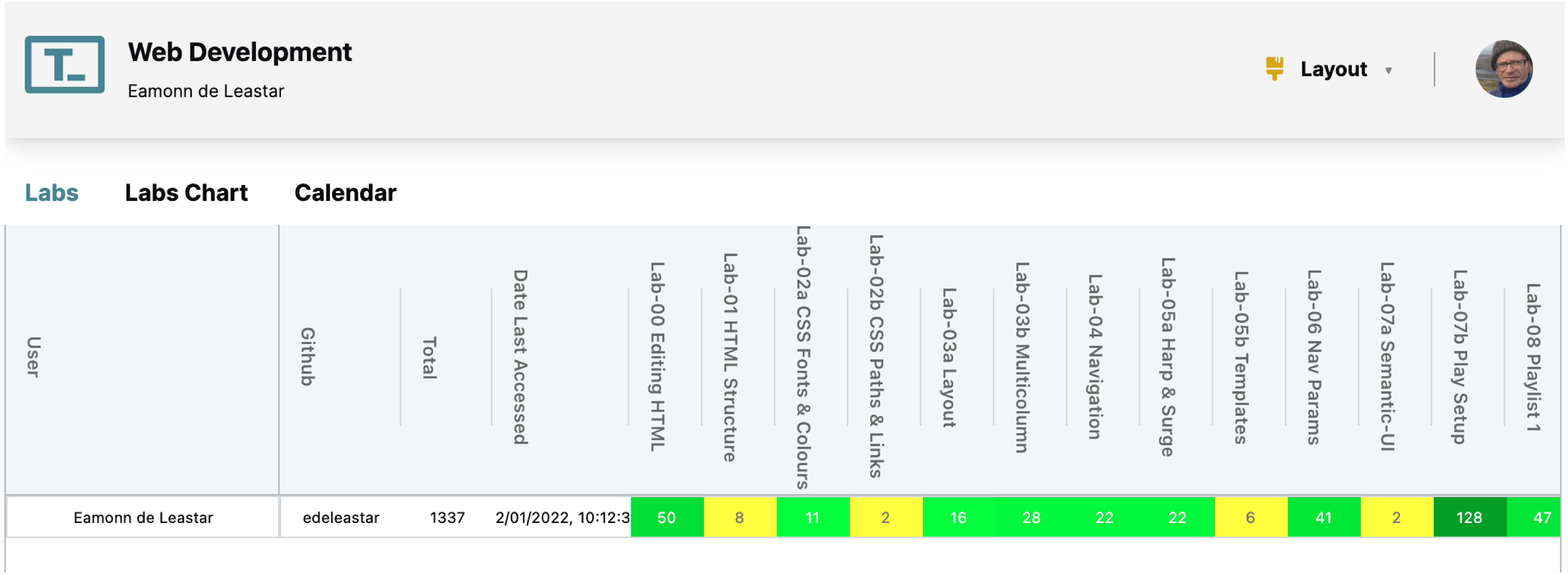
You will be asked to authenticate again in seven days time.

Proceed to Course

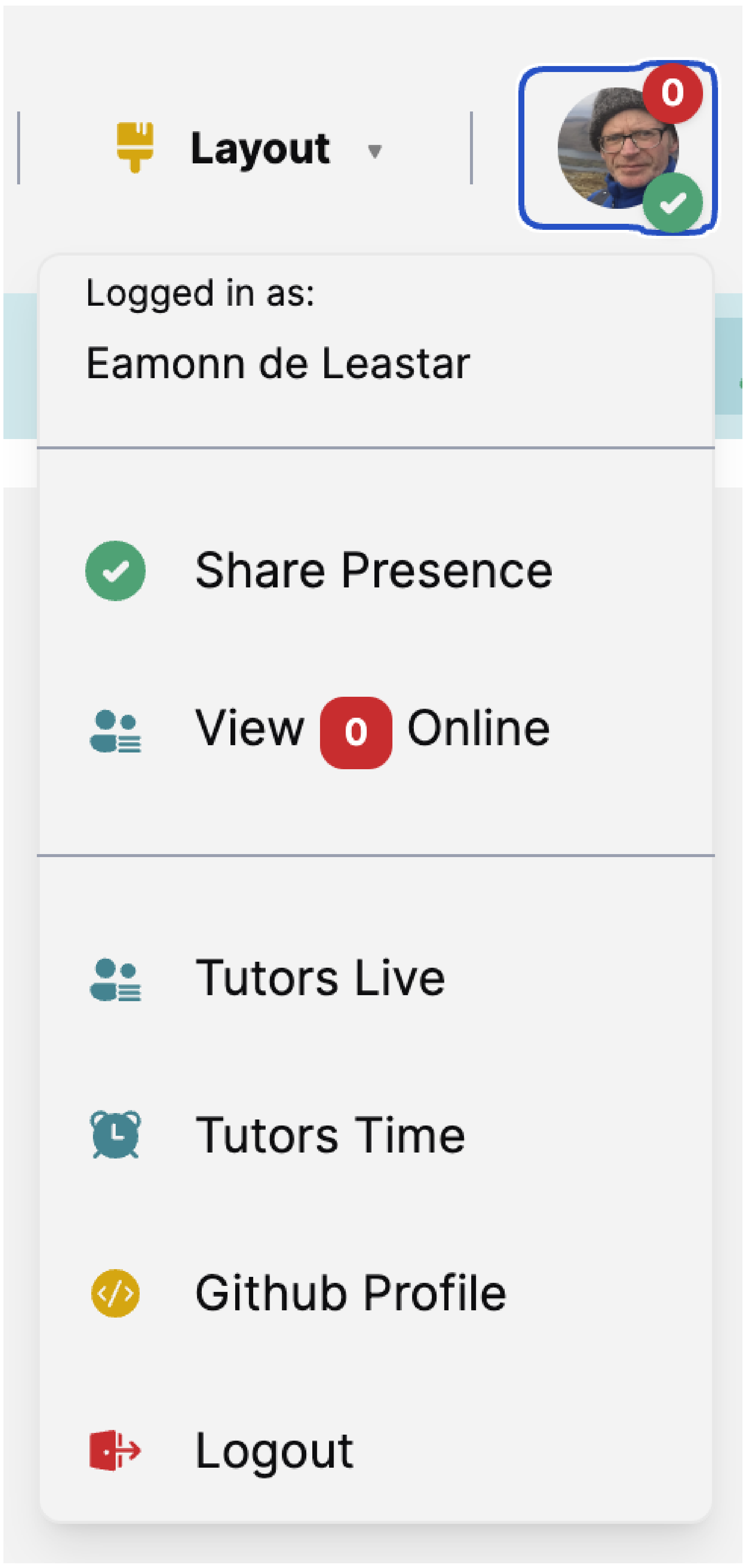
Github Profile



TutorsTime - Labs

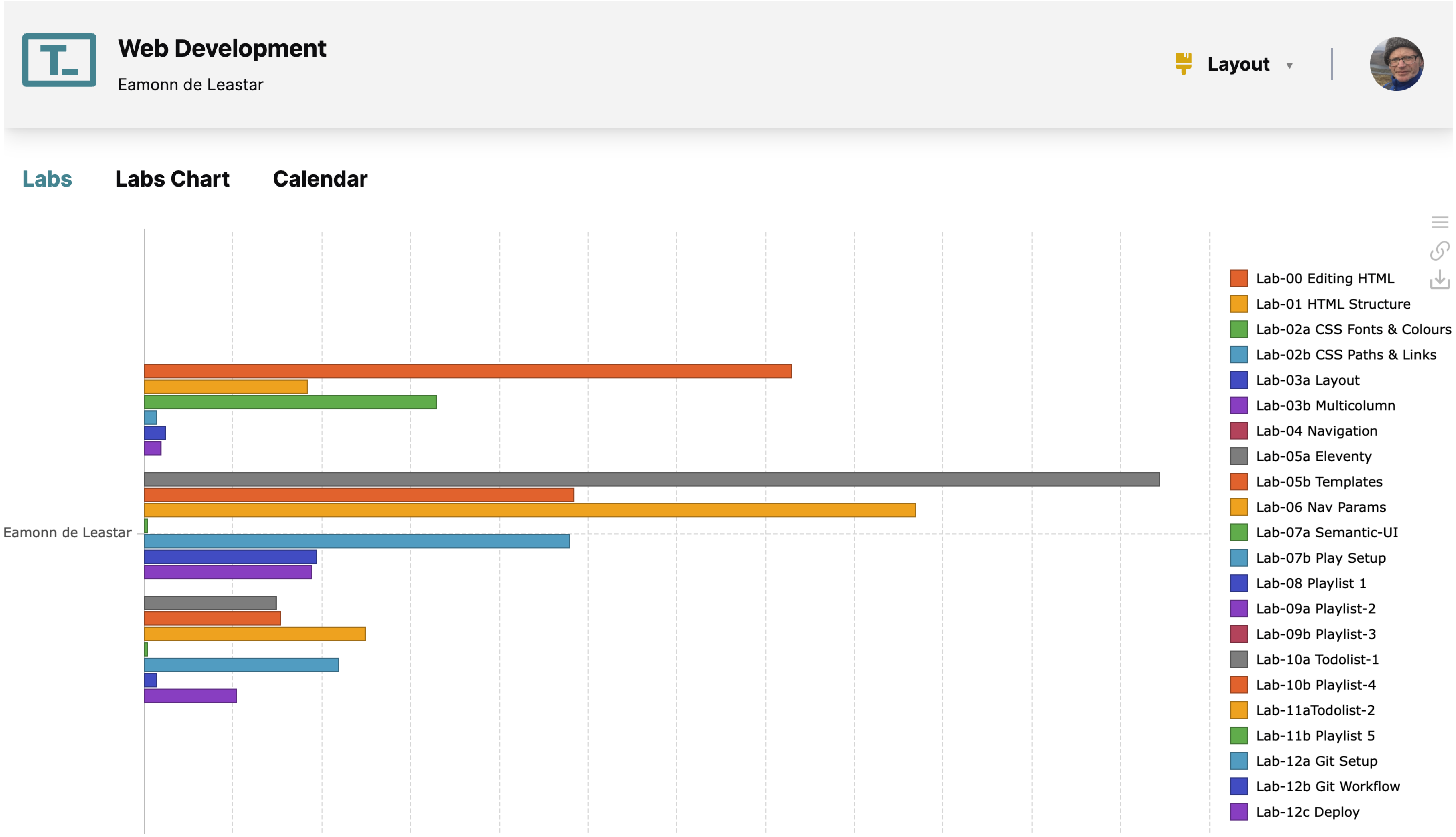


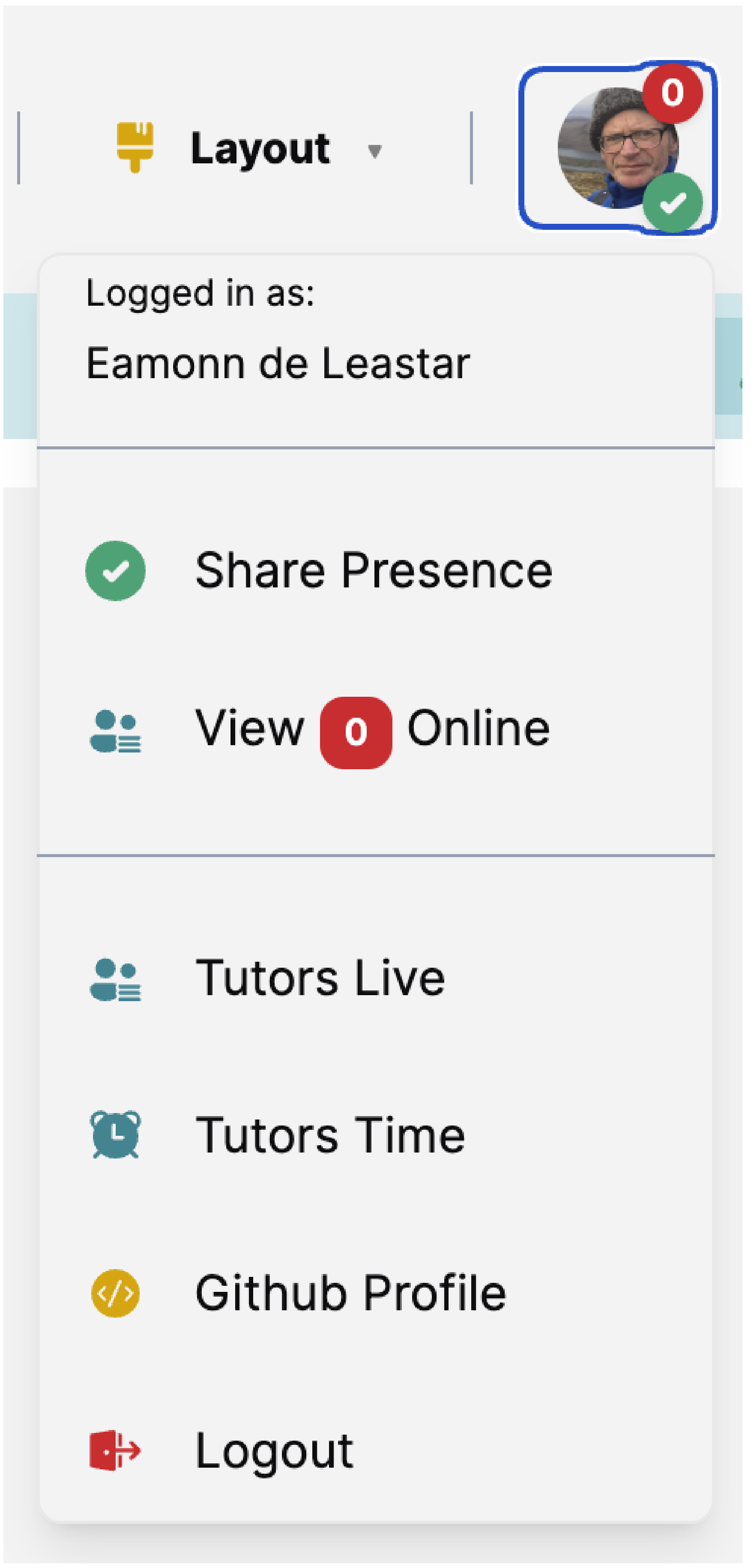
User	Github	Total	Date Last Accessed	Lab-00 Editing HTML	Lab-01 HTML Structure	Lab-02a CSS Fonts & Colours	Lab-02b CSS Paths & Links	Lab-03a Layout	Lab-03b Multicolumn	Lab-04 Navigation	Lab-05a Harp & Surge	Lab-05b Templates	Lab-06 Nav Params	Lab-07a Semantic-UI	Lab-07b Play Setup	Lab-08 Playlist 1
Eamonn de Leastar	edeleastar	1337	2/01/2022, 10:12:3	50	8	11	2	16	28	22	22	6	41	2	128	47



Number of minutes the Each lab has been active in your browser (estimate)

TutorsTime - Chart





At a glance comparative time spent on each labs

TutorsTime - Calendar

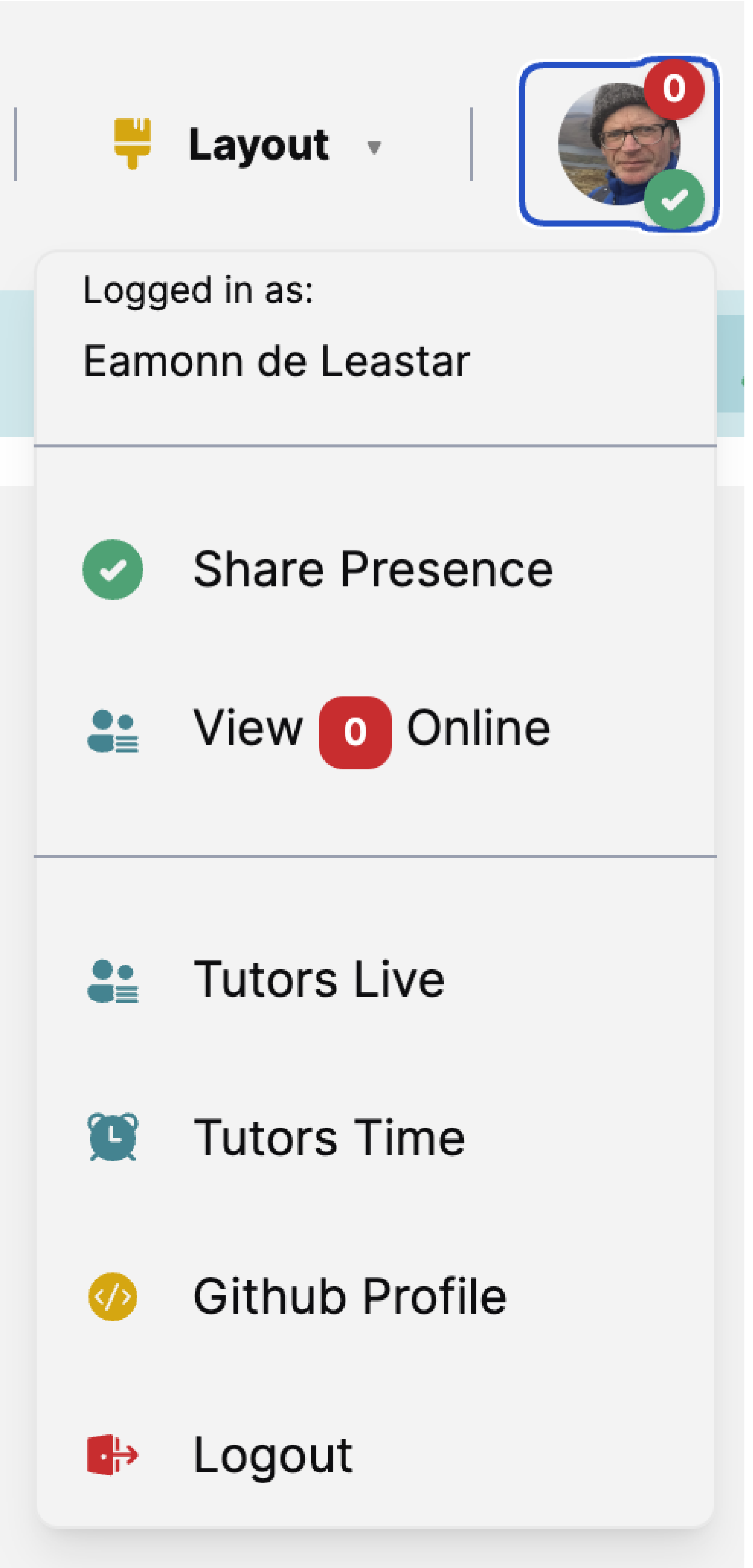
Web Development

Eamonn de Leastar

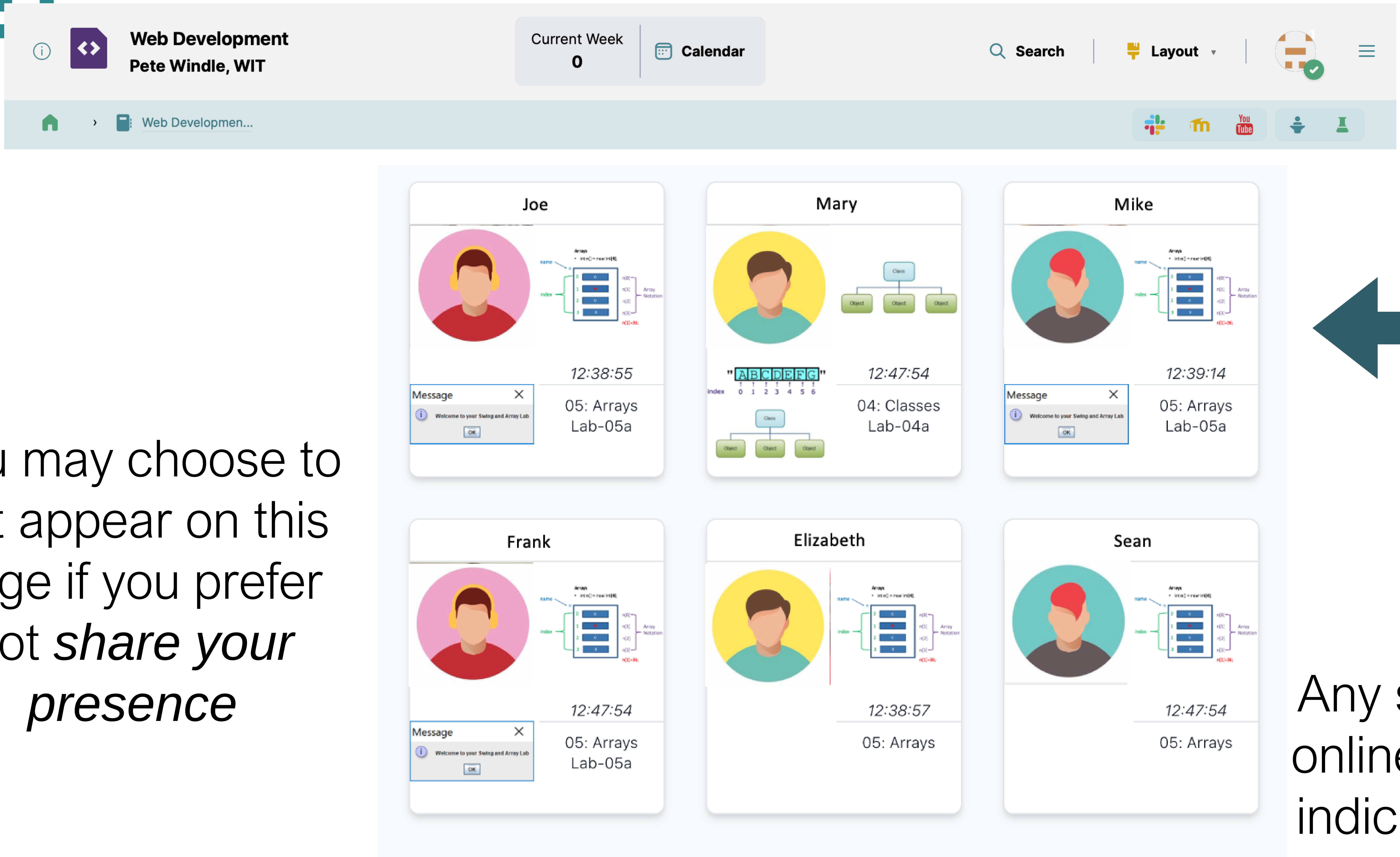
Layout

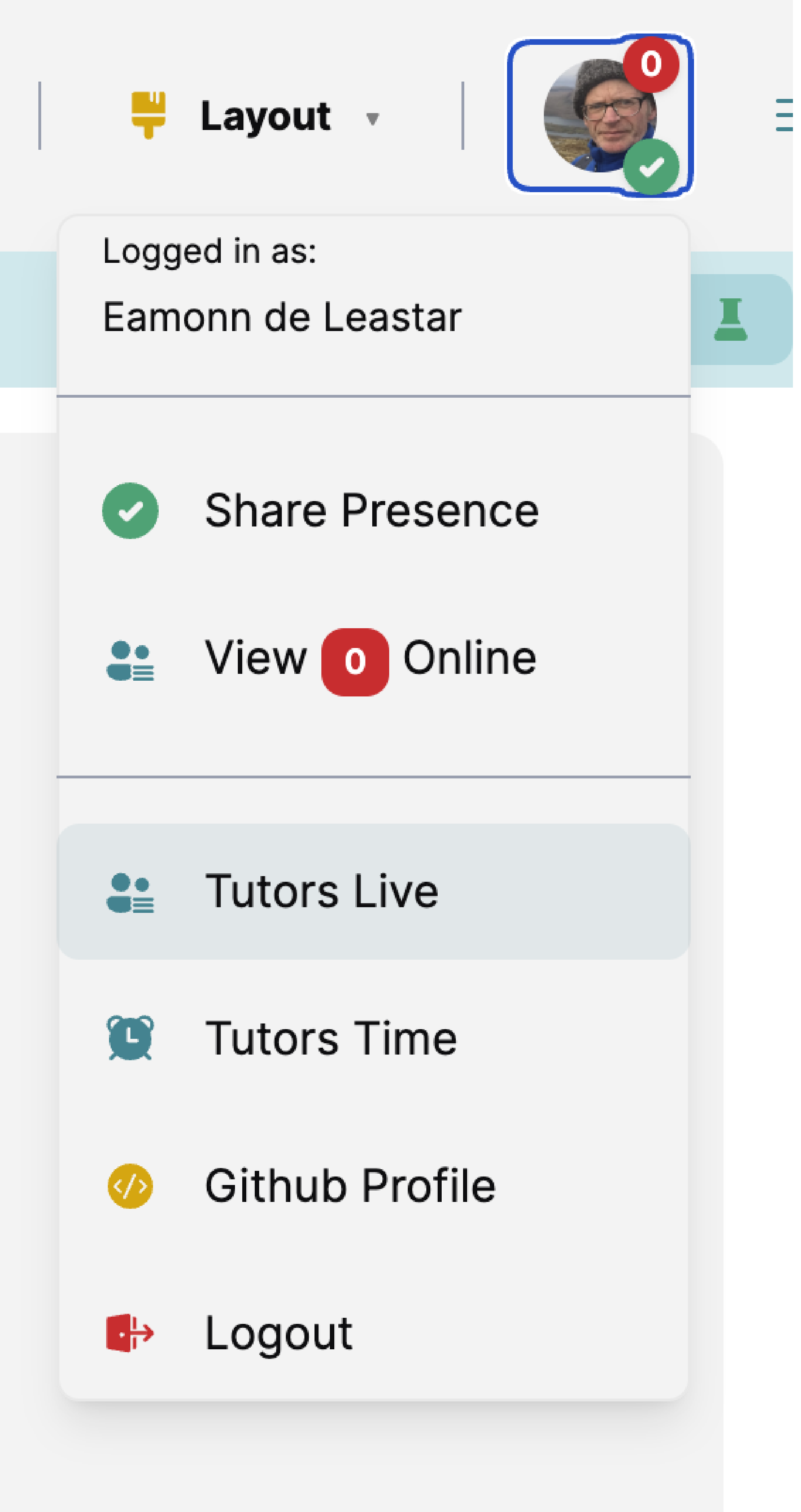
Labs | Labs Chart | Calendar

User	Github	Day	-1: Jan	0: Jan	1: Jan	2: Jan	3: Feb	Reading 1: Feb	4: Feb	5: Feb	6: Mar	Reading 2: Mar	7: Mar	8: Mar	9: Mar	Reading 3: Apr
Eamonn de Leastar																
		Mon	1	4		6	2	109		4	21	2		61	5	18
		Tue	57		12	44	54			46	46					42
		Wed			164	64	6			25	82	51		76	53	
		Thur	2		505	70	327	10	157	192	380	156	258	546	62	13
		Fri	4	1	38	11	19		2	122	302	18	9	111	10	2
		Sat	36		14	7	4				8				138	41
		Sun			16	17			1			6			14	1



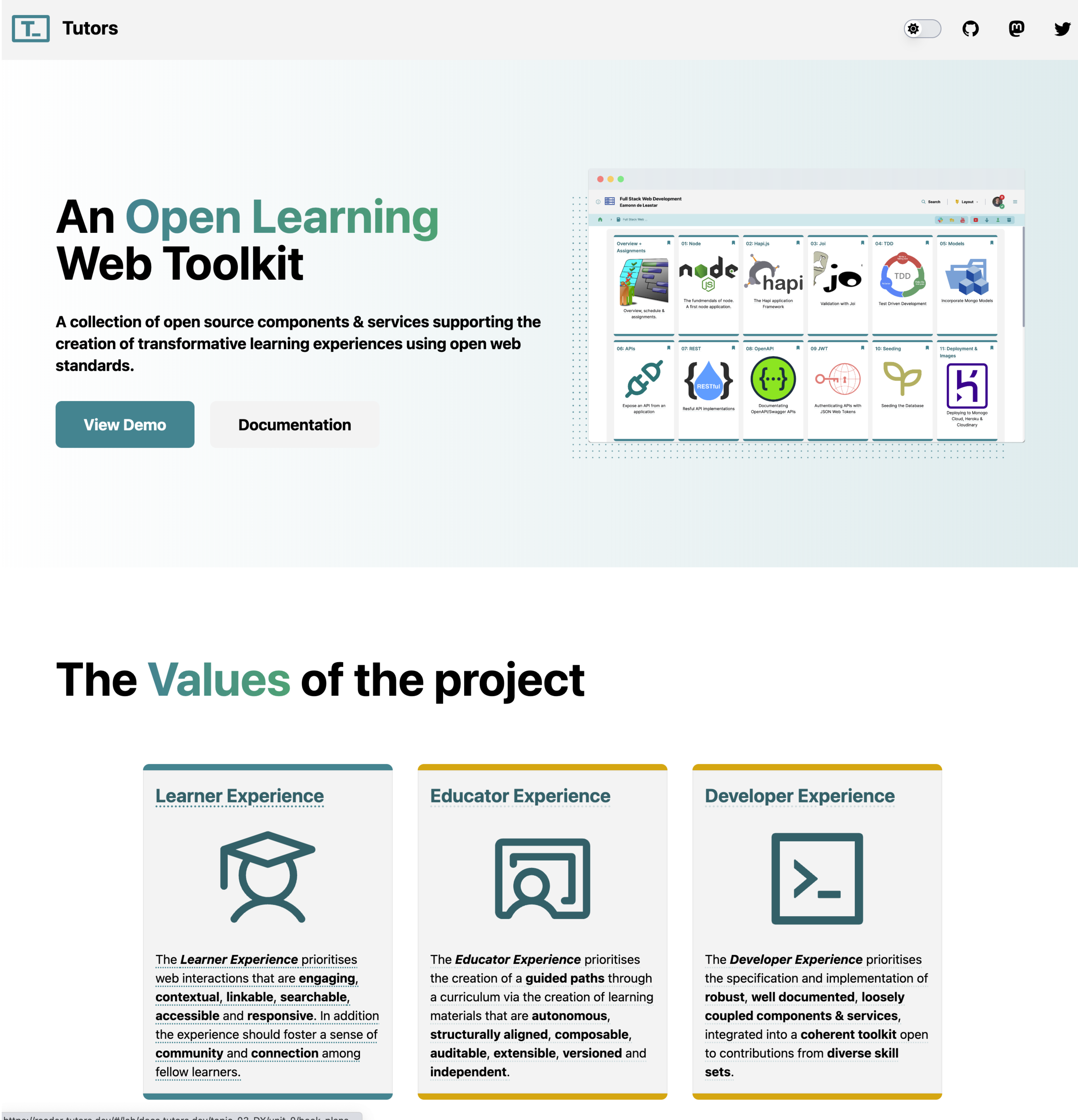
Time online each week of the semester





You may choose to not appear on this page if you prefer not *share your presence*

Any students currently online will appear here, indicating the topic/lab they may be working on https://tutors.dev



https://reader.tutors.dev/course/tutors-project-site